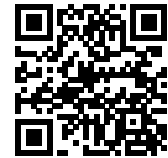
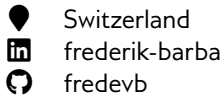


Frederik Barba



Portfolio & Projects

<https://fredevb.github.io/portfolio>

EDUCATION

ETH Zürich **Switzerland**
MSc Computer Science 2024 – Present

- **Specialization:** Machine Intelligence

Technical University of Denmark **Denmark**
BSc Computer Science 2019 – 2023

Caltech **USA**
Exchange program 2022

Nanyang Technological University **Singapore**
Exchange program 2022

INDUSTRY EXPERIENCE

CERN **Switzerland & France**
Software Engineer 2024

- Integrated two hardware-accelerated libraries (*covfie* & *tracc*) into the *ACTS* framework by designing low-overhead data interfaces in C++ to support GPU-accelerated particle track reconstruction.
- Benchmarked the new reconstruction pipeline against the legacy codebase across 50+ accuracy metrics, analyzing intermediate steps and single vs. double precision.

Summer Intern 2023

- Extended *Detray*, an *ACTS* parallelization R&D project, with a C++ visualization and debugging module, exposing two critical particle detector bugs immediately upon deployment.
- Completed curriculum in applied/theoretical physics, GPU programming, and neural networks.

Netcompany **Denmark**
IT Consultant 2023

- Engineered and tested a full-stack government portal (C#, SQL, Angular) handling Brexit transitions for the fishing industry.

Nykredit **Denmark**
Software Engineer 2021 – 2022

- Independently built RPA software (C#) to automate the repair of malformed documents, annually processing 3,000+ files and eliminating 100+ hours of manual workflows.
- Designed UI-driven tools (.NET, SQL) to migrate legacy MS Access databases to Oracle DB.

SKILLS

- **Technologies:** C/C++, Python, C#, Rust, Java, SQL, Linux, Bash, CMake, HDL, Assembly, PyTorch, OpenCV, ROS2, Docker, Unity, Ghidra, git.

RESEARCH EXPERIENCE

University of Tokyo **Japan**
Summer Researcher (Quantum Sensing Lab) 2025

- Programmed Geant4-based C++ simulations of an entangled-photon Compton scattering experiment, verifying theoretical predictions and guiding next-generation PET design.
- Reduced data generation runtime from 2 days to 15 seconds by developing an idealized simulation variant (37,000+ events/sec) for rapid prototyping.

Yonsei University **South Korea**
Summer Researcher (Quantum Computing Lab) 2022

- Evaluated Quantum GAN architectures, analyzing theoretical performance and structural trade-offs for the lab.

TEACHING & VOLUNTEERING

Volunteers Initiative Nepal **Nepal**
Computer Science Teacher 2023 – 2024

- Taught Python programming and foundational computer science to 100+ children from marginalized communities.

Student Organization **Denmark**
Math Tutor 2020 – 2021

- Mentored university students for the course *Advanced Engineering Mathematics*, guiding them through weekly problem sets.

SELECTED COURSEWORK

- **AI & Algorithms:** Probabilistic Algorithms, Deep Learning, Reinforcement Learning, Probabilistic AI, Computer Vision, Robotics.
- **Systems & Databases:** Advanced Systems Lab, Computer Architecture, Cloud Computing Architecture, Big Data, Operating Systems.
- **Networks & Security:** Distributed Computing, Computer Networks, Information Security Lab.

BACKGROUND & HOBBIES

Grew up attending international schools across Europe, including four years each in Portugal and Austria. Outside of work, usually found tackling a new hiking trail or organizing board game nights.